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Bounds for the discrete Bak-Sneppen model

Bachelor's Project Applied Mathematics

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Abstract

We study the discrete version of the Bak-Sneppen model, where we have N species arranged in a 1-D ring formation, each with corresponding fitness values, either 0 or 1. At each step a randomly chosen zero fitness species (or one fitness in the all one's case) as well as its 2 neighbouring species on the ring are updated with new fitness values sampled from a Bernoulli(p) random variable. We then analyse the model's long term behaviour. The main objectives of this paper will be to present the proof from [9] showing that for p above a certain critical threshold p_c , nearly every species has fitness 1 in the stationary distribution as $N \rightarrow \infty$ and to adapt the approach of [7] by showing that for small p , a positive density of zeros remains, through a comparison with oriented percolation, from which we establish a numerical lower bound for the critical threshold. Finally we will run simulations to support these results as well as obtain a numerical estimate for p_c .

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1. Introduction

The Bak-Sneppen model is a well-known interacting particle system originally proposed by Per Bak and Kim Sneppen in 1993 to model the dynamics of species co-evolution [1]. In its classical formulation, a set of N species with fitness values on a ring evolves by repeatedly replacing the minimum fitness value and its neighbours with new random values. This simple rule leads to complex dynamics that have been extensively studied in probability theory. In this thesis, we analyse a discrete variant of this model, where fitness values are binary, belonging to $\{0, 1\}$. This simplification, while altering the nature of the process, preserves many of the original model's essential features and allows for a more mathematically rigorous analysis.

The central question in the discrete Bak-Sneppen model is the value of a critical p_c for the Bernoulli parameter p , which is the probability of a site receiving a one fitness when subjected to an update. For $p < p_c$, zeros are expected to survive indefinitely in the system and with a positive density. Conversely, for $p > p_c$, the system is predicted to converge to an all-ones state i.e. the probability of a species having fitness one is equal to one. Proving the existence of such a phase transition and bounding the value of p_c has been a significant challenge, particularly because the model lacks a monotone structure, preventing the use of standard monotone coupling arguments.

Previous work by Volkov [9] has established a rigorous upper bound on p_c , demonstrating that for all p above a certain threshold p^* , the stationary distribution is concentrated on configurations with a density of ones approaching one as the system size tends to infinity. Specifically, it was shown that for $p > p^*$, where p^* is the unique positive real solution of $p^5 + 4p^4 + 2p^3 + 3p^2 = 1$, we have that the stationary probability of a species having fitness one $\pi^{(N)}(p) \rightarrow 1$ as $N \rightarrow \infty$. There are two main goals of this thesis. One is to provide a rigorous proof of this result in an approachable and intuitive way. The second is to prove, using techniques adapted from Popov & Volkov in [7], that zeros survive with positive density for sufficiently small p , thus establishing the existence of an active phase in which the system continuously evolves. Our approach involves representing graphically the discrete Bak-Sneppen dynamics in continuous time and then making a comparison to an oriented percolation process, allowing us to translate known supercriticality and metastability results of the percolation model into the persistence of zeros in the original system.

There will be three main contributions of this thesis. First, we provide a detailed explanation of the computations and steps taken in Volkov's adjusted-diameter method for establishing the rigorous upper bound on p_c . Second, using the graphical representation developed for the discrete Bak-Sneppen dynamics, and partitioning space-time into independent '4-blocks' (following the construction of Popov & Volkov), we couple the process to a two-dimensional oriented percolation model and prove that zeros survive with positive density for sufficiently small p , relying on the known supercriticality bounds of [2]. Third, we complement these theoretical results with numerical simulations, first using numerical optimization to find a lower bound $p_c > 0.00138$ and then using simulations to investigate the stationary behaviour of the system for various system sizes and values of p . This will both verify our results and provide an empirical estimate for the critical value. The thesis concludes with a discussion of the implications of these results and several open problems.

2. Preliminary results

In this section, we discuss some important results for later use.

Markov Chain convergence

A discrete-time Markov chain $(X_t)_{t \geq 0}$ on a countable state space S is a stochastic process satisfying the Markov property

$$\mathbb{P}(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots, X_0) = \mathbb{P}(X_{t+1} = j \mid X_t = i) = p_{ij}.$$

The matrix $P = (p_{ij})_{i,j \in S}$ is called the transition matrix.

Definition 2.0.1 (Stationary Distribution). *A probability distribution $\pi = (\pi_j)_{j \in S}$ on S is called a stationary distribution if*

$$\pi P = \pi.$$

Hence, it is invariant under the progression of time.

Definition 2.0.2 (Irreducibility). *A Markov chain is called irreducible if for every pair of states $i, j \in S$, there exists an integer $t \geq 0$ such that*

$$(P^t)_{ij} > 0.$$

This means that every state can be reached from every other state.

Definition 2.0.3 (Aperiodicity). *Let $i \in S$ be any state. An irreducible Markov chain is called aperiodic if*

$$\gcd\{t \geq 1 : (P^t)_{ii} > 0\} = 1.$$

for some (and hence every) state $i \in S$.

This tells us that there is no fixed cycle length that forces the chain to only return to a state at regular periodic intervals.

Theorem 2.0.4 (Convergence Theorem [6]). *Consider a Markov chain that is irreducible and aperiodic, and let π denote the unique stationary distribution on the state space S . Then*

$$(P^t)_{ij} = \mathbb{P}(X_t = j \mid X_0 = i) \longrightarrow \pi_j, \text{ as } t \rightarrow \infty,$$

for all $i, j \in S$.

The theorem states that, regardless of the initial state i , the distribution of X_t converges to the stationary distribution π over time.

Theorem 2.0.5 (Cesàro Summation Theorem). *Let $(a_t)_{t \geq 1}$ be a real sequence such that $a_t \rightarrow a$ as $t \rightarrow \infty$. Then the Cesàro averages also converge to a , i.e.*

$$\frac{1}{t} \sum_{k=1}^t a_k \longrightarrow a \quad \text{as } t \rightarrow \infty.$$

Oriented percolation

The 2-dimensional oriented site-percolation model is described as follows:

Define the space-time lattice

$$\mathcal{L} := \{(x, m) \in \mathbb{Z}^2 : x + m \text{ is even}\}.$$

The set of oriented edges is

$$\vec{E} := \{(\langle (x, m), (x \pm 1, m + 1) \rangle) : (x, m) \in \mathcal{L}\}.$$

Fix $\phi \in [0, 1]$. Each site $(x, m) \in \mathcal{L}$ is independently declared **open** with probability ϕ or **closed** with probability $1 - \phi$.

An *open oriented path* is a sequence

$$(x_0, m_0) \rightarrow (x_1, m_1) \rightarrow \cdots \rightarrow (x_k, m_k),$$

such that each site is open, and $m_0 < m_1 < \cdots < m_k$.

We write

$$(x, m) \rightarrow (y, n)$$

if there exists an open oriented path from (x, m) to (y, n) .

The *cluster* of (x, m) is given by (see figure 2.1)

$$C_{(x,m)} := \{(y, n) \in \mathcal{L} : (x, m) \rightarrow (y, n)\}.$$

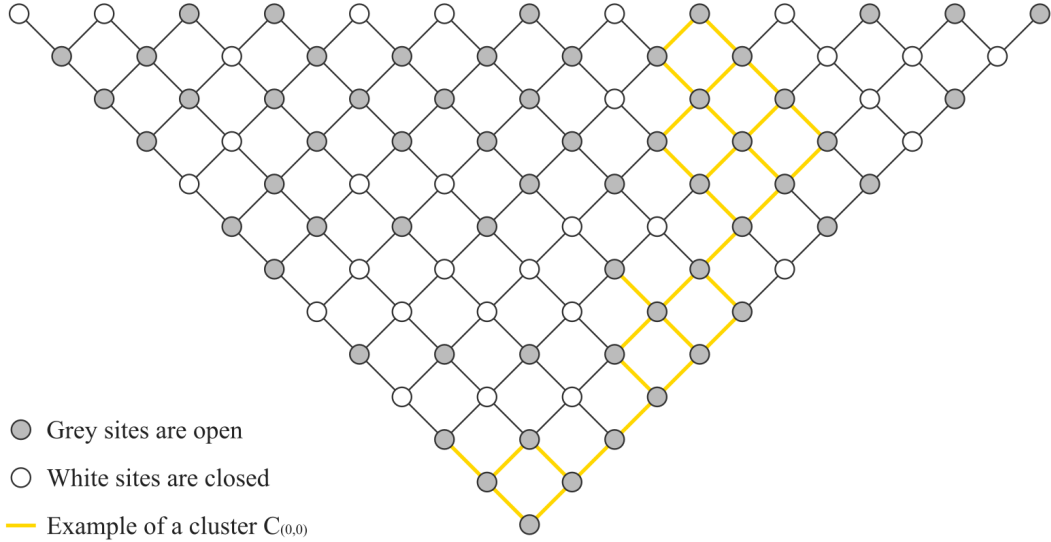


Figure 2.1: Oriented site percolation showing the cluster $C_{(0,0)}$.

We then define the probability that the cluster of the origin is infinite (in this case we say the model percolates) as

$$\theta(\phi) := \mathbb{P}_\phi(|C_{(0,0)}| = \infty)$$

The critical parameter is defined as

$$\phi_c := \inf\{\phi \in [0, 1] : \theta(\phi) > 0\}$$

where if $\phi > \phi_c$, we say that we are in the supercritical regime, and subcritical if $\phi < \phi_c$. We are specifically interested in a rigorous upper bound on this critical probability. From [2] it has been proven that

$$\phi_c < 0.72599.$$

Shape theorem and metastability results

Here, we will state some useful theorems taken from [3, 4, 5, 8], rewritten for our purposes. First, we define

$$\begin{aligned} r_n &= \sup\{x : (x, n) \in \mathcal{L}, (0, 0) \rightarrow (x, n)\} \\ l_n &= \inf\{x : (x, n) \in \mathcal{L}, (0, 0) \rightarrow (x, n)\} \end{aligned}$$

Theorem 2.0.6 (Shape theorem). *On the event $\{|C_{(0,0)}| = \infty\}$, there exists a deterministic constant $\alpha(\phi) > 0$ such that $\mathbb{P}_\phi$ -almost surely*

$$\frac{r_n}{n} \rightarrow \alpha(\phi), \quad \frac{l_n}{n} \rightarrow -\alpha(\phi).$$

We see that given an infinite percolation occurs, the cluster of the origin expands at a stable rate in a cone-like formation and thus, is asymptotically contained in the interval $[-\alpha(\phi)n, \alpha(\phi)n]$.

Now, for **all** the following, suppose that we are in the supercritical regime, i.e. $\phi > \phi_c$. We define for some $A \subseteq \mathbb{Z}$

$$\xi_n^A := \{x : (x, n) \in \mathcal{L}, \exists a \in A \text{ such that } (a, 0) \rightarrow (x, n)\}.$$

Theorem 2.0.7. *The distribution of $\xi_n^{\mathbb{Z}}$ converges to $\bar{\nu}$ as $n \rightarrow \infty$ ($\bar{\nu} = \bar{\nu}(\phi)$). $\bar{\nu}$ is called the upper invariant law of the process. The density of $\bar{\nu}$ is given by*

$$\bar{\nu}(x) = \bar{\nu}(\xi : x \in \xi) = \mathbb{P}_\phi(|C_{(0,0)}| = \infty) > 0.$$

This is translation-invariant, i.e. $\bar{\nu}(x) = \bar{\nu}(y)$ for any $x, y \in \mathbb{Z}$.

Hence, the process converges to an equilibrium in which, for a given site, there is an open path from the initial configuration with positive probability.

Theorem 2.0.8. *$\mathbb{P}_\phi$ -almost surely, for every $\varepsilon > 0$, there exists some $N = N(\varepsilon) < \infty$ such that for all $n \geq N$,*

$$\begin{aligned} &\text{for all } x \in [-\alpha(\phi)(1 - \varepsilon)n, \alpha(\phi)(1 - \varepsilon)n] \text{ we have} \\ &x \in \xi_n^{(0,0)} \iff x \in \xi_n^{\mathbb{Z}}. \end{aligned}$$

Hence, inside a slightly smaller cone than in shape theorem, the processes started from a single site and from all sites eventually has the same distribution.

Theorem 2.0.9. *There exists $\epsilon = \epsilon(\phi) > 0$ such that for all n*

$$\mathbb{P}_\phi(\xi_n^{(0,0)} \neq \emptyset \mid |C_{(0,0)}| < \infty) \leq e^{-\epsilon n}.$$

Thus, conditioned on eventual extinction, the probability of long survival before extinction decays exponentially.

For some $N \in \mathbb{N}$, let ${}_N\xi_n$ denote the process restricted to a finite interval $[-N, N]$.

The following results come from metastability. This is the phenomenon where a system is in a stable and active state for a very long time before eventually transitioning to extinction. We define the extinction time

$$T_N := \inf\{n > 0 : {}_N\xi_n = \emptyset\}.$$

Let $\beta_N = \beta_N(\phi)$ be the unique value such that

$$\mathbb{P}_\phi(T_N > \beta_N) = e^{-1}.$$

Theorem 2.0.10. *As $N \rightarrow \infty$,*

$$\frac{T_N}{\beta_N} \xrightarrow{d} \text{Exp}(1),$$

where $\text{Exp}(1)$ denotes the exponential distribution with unit mean.

The following theorem tells us that starting with a sufficiently large initial configuration, the process will with arbitrarily high probability have a positive fraction of occupied sites inside any fixed interval after a long time.

Theorem 2.0.11. *For all $\epsilon > 0$, there exists $b = b(\epsilon)$ and $N = N(\epsilon) > b$ such that*

$$\mathbb{P}_\phi(|{}_N\xi_n \cap [-b, b]|/2b < \bar{\nu}(0)/2) < \epsilon.$$

Other results

The following lemmas are taken from [9] and used in the rigorous upper bound proof.

Lemma 1. *Consider a non-negative real-valued stochastic process Y_t , $t = 0, 1, 2, \dots$ for which there exist constants $C, b, \epsilon > 0$ such that the differences $\Delta_t = Y_t - Y_{t-1}$ satisfy*

- (a) $\Delta_{t+1} \leq b$ for all t ;
- (b) $\mathbb{E}[\Delta_{t+1} \cdot \mathbf{1}_{\Delta_{t+1} \geq -b} \mid Y_t] \leq -\epsilon$ whenever $Y_t \geq C$.

Then there exists a finite constant $R_m = R_m(C, b, \epsilon)$ such that for every $m = 1, 2, \dots$

$$\limsup_{t \rightarrow \infty} \mathbb{E}(Y_t^m) \leq R_m.$$

Lemma 2. *Consider a non-negative real-valued stochastic process M_t , $t = 0, 1, 2, \dots$, and a strictly increasing sequence of positive integer stopping times τ_k , $k = 0, 1, 2, \dots$, with $\tau_0 = 0$. Suppose there exist constants $C, b, r, \epsilon > 0$ such that*

- (a) $M_{t+1} - M_t \leq b$ for all t ;
- (b) $\mathbb{E}[(M_{\tau_{k+1}} - M_{\tau_k}) \cdot \mathbf{1}_{M_{\tau_{k+1}} - M_{\tau_k} \geq -b} \mid M_{\tau_k}] \leq -\epsilon$ whenever $M_{\tau_k} \geq C$;
- (c) $M_{t+1} \leq M_t$ for all $t \notin \{\tau_k : k \geq 0\}$;
- (d) $\mathbb{E}(\tau_{k+1} - \tau_k \mid M_{\tau_k}) \leq r(1 + M_{\tau_k})$.

Then

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E} M_t \leq \tilde{R},$$

where $\tilde{R} = \tilde{R}(C, b, r, \epsilon) < \infty$ does not depend on the initial condition.

3. Bak-Sneppen model

Generalized Bak-Sneppen model

We first want to introduce a general case of the Bak-Sneppen model. Suppose that we have $N \geq 3$ vertices in a connected graph $G = (V, E)$, where $V = \{1, 2, \dots, N\}$, $|V| = N$.

For each vertex $i \in V$, define its neighborhood by

$$\mathcal{N}(i) := \{j \in V : (i, j) \in E\}.$$

The model evolves in discrete time steps $t = 0, 1, 2, \dots$ where every vertex $i \in V$ is assigned a certain **fitness value**

$$X_i(t) \in [0, 1].$$

The Bak-Sneppen process $\{X(t)\}_{t \geq 0}$ forms a Markov chain on the state space $[0, 1]^N$ where the state of the chain at time t is $X(t) = (X_i(t))_{i \in V}$.

We assume that initially $\{X_i(0)\}_{i \in V}$ are independent and uniformly distributed on $[0, 1]$, so that ties occur with probability zero, and thus there is a unique minimum fitness value.

At each time t , choose index $j = j(t)$ such that $X_j(t) = \min_{i \in V} X_i(t)$. Then we update as follows

$$X_i(t+1) = \begin{cases} U_i(t+1), & i \in \{j\} \cup \mathcal{N}(j) \\ X_i(t), & i \notin \{j\} \cup \mathcal{N}(j) \end{cases}$$

where for all $t = 0, 1, 2, \dots$

$$U_i(t+1) \sim \text{Unif}(0, 1).$$

are i.i.d. random variables. This defines the Bak-Sneppen dynamics on an arbitrary graph G .

Bak-Sneppen model on a ring

In the classical Bak-Sneppen model, the underlying graph is a ring C_N , consisting of N vertices arranged on a circumference.

Each vertex has exactly two nearest neighbours. In this model, we impose **periodic boundary conditions**, thus the neighbourhood of any vertex $i \in V$ is defined by

$$\mathcal{N}(i) = \begin{cases} \{i-1, i+1\}, & i \in \{2, \dots, N-1\}, \\ \{N, 2\}, & i = 1, \\ \{N-1, 1\}, & i = N. \end{cases}$$

Discrete Bak-Sneppen model

The subsequent sections work exclusively with the discrete Bak-Sneppen model. In this model, the fitness values are no longer continuous in $[0, 1]$, but instead take values in binary state space

$$X_i(t) \in \{0, 1\}, \quad i \in V, \quad t = 0, 1, 2, \dots$$

so that the process becomes a Markov chain on $\{0, 1\}^N$.

Instead of drawing new fitness values from the uniform distribution, the updated values are now i.i.d. Bernoulli random variables:

$$U_i(t+1) \sim \text{Bernoulli}(p), \quad p \in [0, 1].$$

In this case the minimum value is typically 0 and clearly it no longer must be unique in general. We therefore define

$$I(t) := \{i \in V : X_i(t) = 0\}.$$

If $I(t) \neq \emptyset$, we choose the update index $j = j(t)$ uniformly among all vertices in $I(t)$:

$$j(t) \sim \text{Uniform}(I(t))$$

and if all vertices are equal to 1, that is, if $I(t) = \emptyset$, then

$$j(t) \sim \text{Uniform}(V).$$

Critical parameter

We define the critical parameter by

$$p_c := \sup\{p \in [0, 1] : \theta_{BS}(p) > 0\},$$

where $\theta_{BS}(p) := \lim_{N \rightarrow \infty} \mathbb{P}(X_i(t) = 0 \text{ in the stationary distribution})$ denotes the probability of survival of zeros in the limit of infinitely many vertices (stationary distribution defined 4.1).

For p_c to be well-defined, we must have that $\theta(p)$ is non-increasing in p (this should intuitively be the case). Observe that the discrete Bak-Sneppen process is not monotone with respect to the natural partial order on configurations. For example, under the same update, the configurations $(0, 1, 0)$ and $(0, 0, 0)$ can update to $(0, 1, 0)$ and $(1, 1, 1)$ respectively, so the ordering is not preserved. For this reason, standard monotonicity-based coupling and stochastic domination arguments used to establish the existence of a sharp critical parameter as used in oriented percolation are not directly available here. For this reason, we will instead make the assumption that p_c is well-defined.

We then have that p_c acts as a critical threshold that separates a regime in which zeros survive from one in which they die out. In other terms for $p < p_c$, we assume that in the stationary distribution each vertex has zero fitness with positive probability, while for $p > p_c$, almost all vertices have fitness equal to one.

4. Rigorous upper bound for critical value

Since $\{X(t)\}_{t \geq 0}$ is irreducible and aperiodic on the finite state space $\{0, 1\}^N$, it admits a unique stationary distribution π and we converge to it as $t \rightarrow \infty$ for any initial condition by the Markov chain convergence theorem 2.0.4.

We may therefore define

$$\pi^{(N)}(p) = \pi_{N,p}(X : X_i = 1) = \lim_{t \rightarrow \infty} \mathbb{P}(X_i(t) = 1) \quad (4.1)$$

as the stationary probability that vertex i is in state 1 in the discrete Bak-Sneppen model on the ring C_N with N vertices and Bernoulli parameter p . Note that, by symmetry, this is independent of index i . We may also observe that $1 - \pi^{(N)}(p) \rightarrow \theta_{BS}(p)$ as $N \rightarrow \infty$.

In this section we aim to rigorously show that

$$\pi^{(N)}(p) \rightarrow 1 \text{ as } N \rightarrow \infty \text{ when } p > p^* = 0.45\dots, \quad (4.2)$$

i.e. that in the stationary distribution, as the number of vertices grows large, almost every vertex is in state 1. More specifically, p^* is the only real positive solution of the equation $p^5 + 4p^4 + 2p^3 + 3p^2 = 1$.

Hence, we obtain an upper bound for our critical parameter, $p_c \leq p^*$.

Proof setup

Define the total number of zeros at time t

$$K_t = |\{i : X_i(t) = 0\}|,$$

and for a given configuration $X_1(t), \dots, X_N(t)$, the minimal spanning arc as a subset of indices of $\{1, 2, \dots, N\}$ taken modulo N and given by

$$S_t = \{i, i+1, \dots, j-1, j\},$$

with the following properties:

1. $X_i(t) = 1$ for all $i \notin S_t$
2. S_t has the smallest cardinality amongst all such subsets.

Thus, the minimal spanning arc is the shortest consecutive sequence of vertices on the ring C_N that contains all the zeros. Let the pair (a_t, b_t) be the pair with the first and last index of S_t , with $a_t = b_t = 0$ if $K_t = N$ or $K_t = 0$. If there is more than one such sets, S_t is chosen as the one with the smallest starting index a_t . This makes S_t a deterministic function of $X(t)$ and differs from [9] which proposes an arbitrary choice of S_t that leads to the adjusted diameter M_t in the following definition not being defined.

We then denote the length of such an arc by

$$D_t = |S_t| \geq K_t.$$

Definition (Adjusted diameter). Suppose $D_t \geq 6$. For some constant $\beta \in (0, 1/2)$, the adjusted diameter of the configuration is defined as

$$M_t = D_t + 1 - \beta \left(\mathbf{1}_{X_{a_t+1}(t)=0} + \mathbf{1}_{X_{b_t-1}(t)=0} \right) \in (D_t, D_t + 1]$$

where $\mathbf{1}_{(\cdot)}$ denotes the indicator function. We set $M_t = 0$ whenever $D_t < 6$.

The adjusted diameter is a slight refinement of the arc length that penalises configurations where the vertices just inside the endpoints of the arc are zero. It will later provide us with a better bound on p_c compared to if we had simply used the standard diameter.

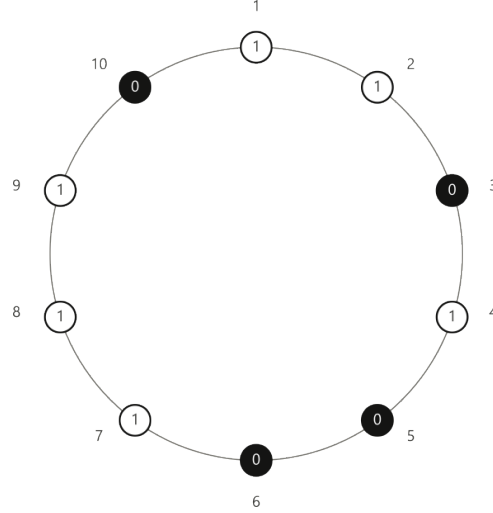


Figure 4.1: Bak-Sneppen model on C_{10} where the black vertices denote our zeros and white denotes our ones. In this state, our minimal spanning arc is given by $S_t = \{10, 1, 2, 3, 4, 5, 6\}$ and hence $D_t = 7$. The vertices just inside the endpoints $\{1, 5\}$ have one and zero fitness respectively and thus we have adjusted diameter $M_t = 8 - \beta$.

Observe that for a new configuration, there are the following possible ways for the minimal spanning arc to update:

1. $S_{t+1} = S_t$;
2. $S_{t+1} = S_t \cup \{a_t - 1\}$;
3. $S_{t+1} = S_t \cup \{b_t + 1\}$;
4. $S_{t+1} \subset S_t$ with $a_{t+1} = a_t + k$ for some $k \geq 1$;
5. $S_{t+1} \subset S_t$ with $b_{t+1} = b_t - k$ for some $k \geq 1$;
6. $S_{t+1} \neq S_t$ with $a_{t+1} = b_t$ and $b_{t+1} = a_t$.

The final case we will call a flip and it can happen when an update occurs at some index $i_t \notin \{a_t, a_t + 1, a_t + 2\} \cup \{b_t - 2, b_t - 1, b_t\}$. For example, consider the configuration

$$X(t) = (1, 0, 1, 0, 0, 0, 1, 0, 1),$$

so that $S_t = \{2, 3, 4, 5, 6, 7, 8\}$. Suppose the zero at index $i_t = 4$ is selected. After updating indices 3, 4, 5 to ones, we obtain

$$X(t+1) = (1, 0, 1, 1, 1, 1, 1, 0, 1),$$

so that $S_{t+1} = \{8, 1, 2\}$ and thus, $(a_{t+1}, b_{t+1}) = (b_t, a_t) = (8, 2)$.

We will now prove 3 useful lemmas about the adjusted diameter, which will form the basis of our proof later.

Lemma 3. *Suppose $D_t \geq 6$. If the endpoints a_t, b_t flip between times t and $t + 1$, then*

$$D_{t+1} \leq D_t - 1 \implies M_{t+1} \leq M_t - (1 - 2\beta) < M_t.$$

Proof: We note that a flip will only occur if the new cardinality of S_t is lower, hence the first inequality. Thus, following from the definition of the adjusted diameter and that $0 < \beta < 1/2$, $M_{t+1} \leq D_{t+1} + 1 \leq D_t \leq M_t - (1 - 2\beta) < M_t$. $\square$

From here, let $a = a_t$ and $b = b_t$. We define the boundary set

$$E_t = \{a, a + 1, a + 2\} \cup \{b - 2, b - 1, b\} \subseteq S_t$$

as the set of the three indices at each endpoint of the minimal spanning arc.

Definition 4.0.1. *Define the sequence of stopping times $(\tau_k)_{k \geq 0}$ by $\tau_0 = 0$ and, for $k \geq 0$,*

$$\tau_{k+1} = \inf\{t > \tau_k : j_t \in E_t\}.$$

Thus, τ_k denotes the k -th time at which the selected index j_t belongs to the boundary set E_t .

Lemma 4. *Suppose $D_t \geq 6$. If the selected index $j_t \notin E_t$, then*

$$M_{t+1} \leq M_t.$$

Proof: In the case of a flip, this result immediately follows from Lemma 3. In the case that a flip does not occur, for $j_t \notin E_t$, the update will neither reach endpoints a and b nor $a + 1$ and $b - 1$, hence $M_{t+1} = M_t$. $\square$

Lemma 5. *Suppose $p > p^*$ and β is given by (4.3), the drift is then given by*

$$\Delta_{t+1} := \mathbb{E}(M_{t+1} - M_t) \text{ where } \Delta_{t+1} \leq 0 \text{ on the event } \{M_t \geq 8\}.$$

Moreover, there exists $\varepsilon = \varepsilon(p) > 0$, depending only on p , such that

$$\Delta_{t+1} \leq -\varepsilon \text{ on the event } \{M_t \geq 8\} \cap \{j_t \in E_t\}.$$

Proof: This proof will be done by a case analysis. We will assume that a flip of indices a and b does not take place since, by Lemma 1, this will always result in a decrease of M_t by at least $1 - 2\beta > 0$.

First, we assume that $D_t \leq N - 2$ and we will look at the remaining two cases later. We note that by symmetry, the drift behaviour at left end point a and right end point b are the same. Hence we can restrict our investigation to the left endpoint a . Our assumption allows to suppose that without loss of generality $a = 2$ and hence, $X_1 = X_2 = 1, X_3 = 0$.

When this is the case at an arbitrary time t , we have four possible configurations:

1. $X = (1, 1, 0, 0, 0, \dots)$
2. $X = (1, 1, 0, 0, 1, \dots)$
3. $X = (1, 1, 0, 1, 0, \dots)$
4. $X = (1, 1, 0, 1, 1, \dots)$

We will now study the 8 possible outcomes for when a triple $(*, 0, *)$ is subjected to the Bernoulli updating step, corresponding to 000, 001, $\dots$, 111, each occurring with a probability depending on p , the probability of choosing a 1.

For these cases, we condition that zeros at indices $i = 3, 4, 5$ can be chosen with equal probability. Looking selectively at an update at one of these indices allows us to find an upper bound on the drift term Δ_{t+1} which we will denote by $T_{00}, T_{01}, T_{10}, T_{11}$ for each respective case. The computations of these terms are quite involved. We omit the full calculations here and instead present only the calculations for T_{10} . The following expressions have been taken from [9].

$$\begin{aligned} T_{00} &= -\frac{1}{3} [p^3 + p^2 + p - 1 + \beta(1 - p)] - \frac{1}{3} [p(p^2 + p + 1) + \beta(p + 1)(1 - p)^2] - \frac{\beta(1 - p)}{3} + \beta \\ T_{01} &= \frac{T_{11}}{2} - \frac{1}{2} [p(p^2 + p + 1) + \beta(1 + p)(1 - p)^2] + \beta \\ T_{10} &= -\frac{1}{2} [p^3 + p^2 + p - 1 + \beta(1 - p)] - \frac{\beta(1 - p)}{2} \\ T_{11} &= -(2p - 1)(p^2 + p + 1) - \beta(1 - p)^2(1 + p) \end{aligned}$$

Computation of T_{10} :

This term corresponds to the maximal expected drift contribution in case (3) of the proof, i.e., for the local configuration at the left endpoint

$$X = (1, 1, 0, 1, 0, \dots),$$

where the minimal spanning arc S_t begins at $a_t = 3$ and the vertices immediately inside the arc are $\{X_{a_t+1}, X_{a_t+2}\} = \{1, 0\}$. In this configuration, only the two zeros at indices a_t and $a_t + 2$ are eligible for selection within the boundary set E_t . Conditioning on each being chosen with equal probability $1/2$, we compute the expected change in the adjusted diameter M_t after one update.

We consider the effect of updating the triple $(X_{a_t-1}, X_{a_t}, X_{a_t+1}) = (1, 0, 1)$ when the selected zero is at position a_t . Denoting unimportant values as $*$ we have the following cases

Configuration	Drift
100 * 0?? ...	$1 - \beta$
101 * 0?? ...	1
11000?? ...	$-\beta$
11010?? ...	0
11100?? ...	$-1 - \beta$
11110?? ...	-2

By summing each contribution to the expectation, we have the expected drift is at most

$$(1 - \beta)(1 - p)^2 + p(1 - p) - \beta p(1 - p)^2 + (-1 - \beta)p^2(1 - p) - 2p^3,$$

When the selected zero is at position $a_t + 2$, we cannot have an increased adjusted diameter and the only case a decrease can happen (of β) is when the one at index $a_t + 1$ is set to a zero. Thus the expected drift at most

$$-\beta(1 - p).$$

Averaging over the two equally likely choices of the selected zero yields

$$\begin{aligned} T_{10} &= \frac{1}{2} \left[(1-\beta)(1-p)^2 + p(1-p) - \beta p(1-p)^2 + (-1-\beta)p^2(1-p) - 2p^3 \right] + \frac{1}{2} [(-\beta)(1-p)] \\ &= -\frac{1}{2} (p^3 + p^2 + p - 1 + \beta(1-p)) - \frac{\beta(1-p)}{2}. \end{aligned}$$

We take this as an inequality because we have taken the outcome for the drift with smallest decrease / largest increase among all possible updates including ignoring flips which we know can result in a large decrease.

It can be shown that

$$\frac{\partial T_{ij}}{\partial p} < 0 \quad \text{for all } i, j \in \{0, 1\}, \quad p \in (0, 1), \quad \beta \in (0, 1/2).$$

For the case of T_{10} ,

$$\frac{\partial T_{10}}{\partial p} = -\frac{1}{2}(3p^2 + 2p + 1) + \beta.$$

Since $3p^2 + 2p + 1 > 1$ and $\beta < 1/2$,

$$\frac{\partial T_{10}}{\partial p} < -\frac{1}{2} + \beta < 0.$$

Hence, if $T(\tilde{p}, \beta) = 0$ for some $\tilde{p}$, where $T(p, \beta) := \max_{i,j \in \{0,1\}} T_{ij}$, then $T(p, \beta) < 0$ for all $p > \tilde{p}$. We now choose β such that the largest possible value of p satisfying $T(p, \beta) \leq 0$ is obtained. By direct comparison of the expressions, we find that the cases T_{00} and T_{11} dominate the remaining two cases at the optimal choice of β , namely

$$\begin{aligned} T_{01} &< T_{00}, & T_{10} &< T_{00}, \\ T_{01} &< T_{11}, & T_{10} &< T_{11}. \end{aligned}$$

Hence,

$$T(p, \beta) = \max\{T_{00}, T_{11}\}.$$

Therefore, the value of β should be chosen such that these two dominant drift terms are balanced. Otherwise, if for example $T_{00} > T_{11}$, then a small variation of β can reduce the maximal drift value. Thus the optimal choice is obtained when

$$T_{00} = T_{11}.$$

Solving these two equations simultaneously gives optimal parameter

$$\beta = -\frac{4p^{*3} + p^{*2} + p^* - 2}{2p^{*3} - 2p^{*2} + 3} = 0.34656 \dots \quad (4.3)$$

where $p = p^*$ is the unique positive real solution of

$$p^5 + 4p^4 + 2p^3 + 3p^2 = 1.$$

At these values, $T_{01} = -0.0669\dots$ and $T_{10} = -0.106\dots$, so that for every $p > p^*$ we have $T(p, \beta) < -\varepsilon(p) < 0$ for some $\varepsilon(p) > 0$ depending only on p .

Finally, the cases $D_t = N$ and $D_t = N - 1$ are handled separately. In the first case $X = (0, 0, \dots, 0)$, and we have

$$\mathbb{E}(M_{t+1} - M_t) = \mathbb{E}(D_{t+1} - D_t) < -[1 - (1 - p)^3] < 0$$

since in both the t and $t + 1$ cases, the vertices just inside the endpoints of the arc must be zero and hence the β terms cancel out, then in every case the diameter must decrease by at least one other than in the all zero case which occurs with probability $(1 - p)^3$.

In the second case, we have two possible sub-cases that can have an increased adjusted diameter within one step:

$$\begin{aligned} X &= (0, \dots, 0, 1, 0, \dots, 0) \\ X &= (0, \dots, 0, 1, 0, 1, 0, \dots, 0) \end{aligned}$$

In the first sub-case, we have if selected index $j_t \in E_t$ then the diameter can increase by one if only a or b is chosen and updated with all zeroes and can decrease by three if on $a + 1$ or $b - 1$ an all ones update occurs. Hence, given that zeroes are uniformly chosen at random, for $p > p_\diamond$

$$\mathbb{E}(M_{t+1} - M_t) < \mathbb{E}(D_{t+1} - D_t) < \frac{1}{3}(+1)p^3 + \frac{1}{3}(-3)(1 - p)^3 < 0.$$

In the second sub-case, we have $\{X_a(t), X_{a+1}(t)\} = \{0, 1\}$ and $\{X_{b-1}(t), X_b(t)\} = \{0, 0\}$. The adjusted diameter can increase only (by $1 + \beta$) if the central zero at a is chosen and updated with all zeroes or (by β) if diameter remains $N - 1$ and indices $a_{t+1} + 1$ and $b_{t+1} + 1$ are updated with zeros otherwise (due to the indicator function term). This is the case if central zero at a is updated with a one and two zeroes or if at b or $a + 2$ the index itself and one other is updated to zero. If any chosen index is updated with all ones, the adjusted diameter will decrease by at least $(2 - \beta)$ and if updated by two adjacent ones it will decrease by at least $(1 - \beta)$. Note that in this case there are only 5 zeros to choose from E_t . Hence, for $p > p_\diamond$ and $\beta = \beta_\diamond$ we have

$$\begin{aligned} \mathbb{E}(M_{t+1} - M_t) &< \frac{1}{5}(1 + \beta)(1 - p)^3 + \frac{1}{5}(+\beta)p(1 - p)^2 + \frac{2}{5}(+\beta)((1 - p)^2 + 2(1 - p)^2) \\ &\quad (- (2 - \beta))(p^3) + (- (1 - \beta))(3p^2) < 0. \end{aligned}$$

Thus proving negative drift. □

Proof

Now we are ready to prove the result of (4.2) by first supposing $p > p^*$.

Taking $C = 8$, $b = 2$, $r = 1$, and $\varepsilon = \varepsilon(p) > 0$ as given in (5), we check that M_t and τ_k satisfy the four conditions of Lemma (2).

- (a) *Bounded jumps.* At each step, the minimal spanning arc S_t can grow by at most one index (either $a_t - 1$ or $b_t + 1$ is added), so $D_{t+1} \leq D_t + 1$. Since $M_t \in (D_t, D_t + 1]$, it follows that $M_{t+1} - M_t \leq 2$, so this condition holds with $b = 2$.
- (b) *Negative drift at stopping times.* By Lemma (5), whenever $M_{\tau_k} \geq 8$ we have

$$\mathbb{E}(M_{\tau_{k+1}} - M_{\tau_k}) \leq -\varepsilon < 0,$$

so condition holds with the same $\varepsilon > 0$.

- (c) *Non-increase between stopping times.* By Lemma (4), whenever $j_t \notin E_t$ we have $M_{t+1} \leq M_t$. Since stopping times τ_k are precisely the times at which $j_t \in E_t$, the process M_t is non-increasing on $t \notin \bigcup_k \{\tau_k\}$.
- (d) *Bounded expected gap between stopping times.* Since all zeros are chosen with equal probability $1/K_t$, the probability that the chosen index j_t falls in E_t satisfies

$$\mathbb{P}(j_t \in E_t) = \frac{|\{i \in E_t : X_i(t) = 0\}|}{K_t} \geq \frac{1}{D_t} \geq \frac{1}{M_t + 1},$$

where the first inequality follows as E_t must contain zeroes and the last inequality uses $D_t \leq M_t + 1$, which follows from the definition $M_t \in (D_t, D_t + 1]$ and $\beta < 1/2$. Hence $\tau_{k+1} - \tau_k$ is stochastically dominated by a geometric random variable with success probability $(M_{\tau_k} + 1)^{-1}$, giving expected gap $\mathbb{E}(\tau_{k+1} - \tau_k) \leq M_{\tau_k} + 1$, i.e. condition holds with $r = 1$.

Thus, all conditions of Lemma (2) are satisfied giving us

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E} M_t \leq \tilde{R},$$

for some finite constant $\tilde{R} = \tilde{R}(C, b, r, \varepsilon)$ that does *not* depend on N .

Since M_t is a deterministic function of $X(t)$ which converges to a stationary distribution, the expectation $\mathbb{E} M_t$ converges as $t \rightarrow \infty$ to the stationary mean

$$\mu^{(N)} := \lim_{t \rightarrow \infty} \mathbb{E} M_t.$$

By the Cesàro summation theorem 2.0.5, since $\mathbb{E} M_t \rightarrow \mu^{(N)}$, its time-average converges to the same limit

$$\mu^{(N)} = \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E} M_t \leq \tilde{R}.$$

Recalling that $K_t \leq D_t \leq M_t + 1$, taking expectations under the stationary distribution and dividing by N gives

$$1 - \pi^{(N)}(p) = \frac{\tilde{\mathbb{E}}[K_t]}{N} \leq \frac{\tilde{\mathbb{E}}[D_t]}{N} \leq \frac{\tilde{\mathbb{E}}[M_t] + 1}{N} = \frac{\mu^{(N)} + 1}{N} \leq \frac{\tilde{R} + 1}{N},$$

where $\tilde{\mathbb{E}}$ denotes the expectation under the stationary distribution.

The right-hand side converges to zero as $N \rightarrow \infty$, and so we have that

$$\pi^{(N)}(p) \rightarrow 1 \quad \text{as } N \rightarrow \infty,$$

which is exactly the statement of (4.2). □

5. Survival of zeros

In this section, our aim is to show how the system evolves for very small p .

We want to see if the zeros will persist as we converge to the stationary distribution. More precisely, the key statement is to show that there exists $p > 0$ such that

$$\limsup_{N \rightarrow \infty} \pi_{N,p}(X : X_i = 1) < 1.$$

That is, even as the system size tends to infinity, the stationary probability that a given site is in state 1 remains uniformly bounded away from 1, meaning that zeros survive with positive probability in equilibrium.

Graphical representation

The discrete Bak-Sneppen model can be represented graphically in space-time as in figure 5.1 for a finite number of vertices. We will consider it on the ring C_N as previously. First, we introduce the notion of continuous time t . At each vertex i we have an independent Poisson clock $(N_i(t))_{t \geq 0}$ of rate $\lambda = 1$. $N_i(t)$ denotes the number of clock rings that have occurred at vertex i up to time t . This is a stochastic process such that:

- $N_i(0) = 0$,
- the increments $N_i(t + s) - N_i(s)$ over disjoint time intervals are independent,
- $N_i(t + s) - N_i(s) \sim \text{Poisson}(t)$ for all $t, s \geq 0$,

At each ringing time t^* of a vertex i (i.e. when $N_i(t)$ jumps by $+1$) with neighbourhood $\mathcal{N}(i) = \{j, k\}$, if $X_i(t^* -) = 1$, then nothing happens. If $X_i(t^* -) = 0$, then we update as follows:

$$X_\ell(t^*) = \begin{cases} U_\ell(t^*), & \ell \in \{i\} \cup \mathcal{N}(i), \\ X_\ell(t^* -), & \text{otherwise.} \end{cases}$$

where $(U_i(t^*), U_j(t^*), U_k(t^*))$ are i.i.d. Bernoulli(p) random variables, independent for each ringing time t^* .

This update rule gives us a continuous-time Markov process $\{X(t)\}_{t \geq 0}$ in which updates occur locally at Poisson times and interactions propagate through C_N . This corresponds to the space-time graphical representation of figure 5.1 where each vertex $i \in \{1, 2, \dots, N - 1\}$ is represented as a discrete point along the spatial axis.

The process obtained by observing the continuous-time process immediately after each successful update is identical to the discrete-time Bak-Sneppen model. The next vertex to ring among the currently zero vertices is uniformly distributed, and the update rule is the same and if no zero exists, a one is chosen uniformly at random instead. Hence the transition probabilities coincide and the process converges to the same stationary distribution. This equivalence justifies the use of the graphical representation in the subsequent analysis.

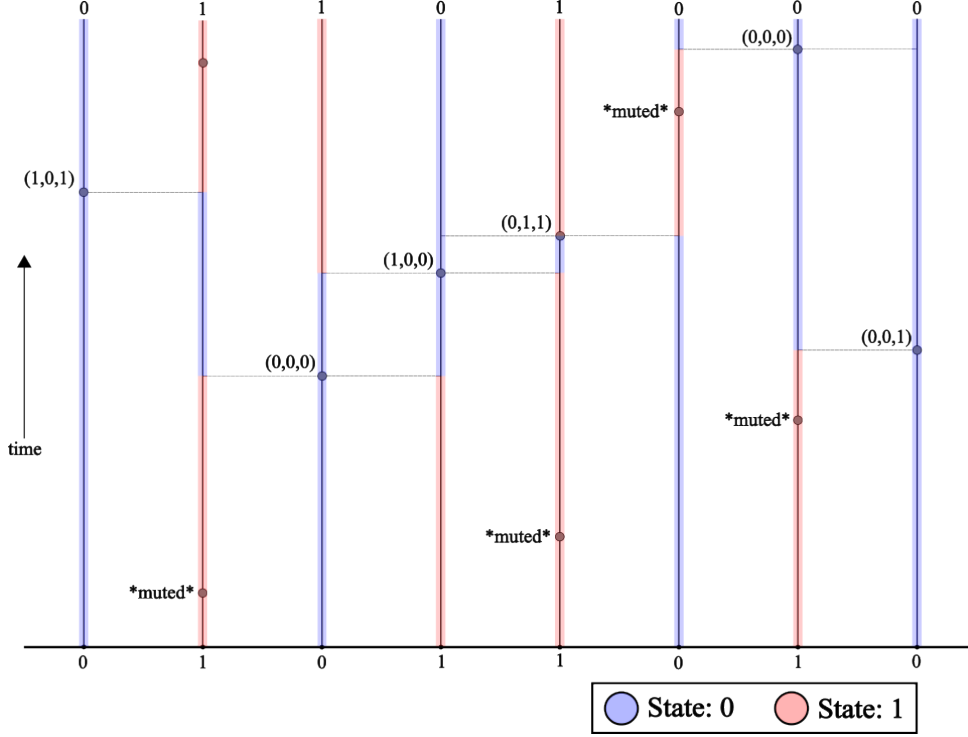


Figure 5.1: The discrete Bak-Sneppen graphical representation using a Poisson clock of rate one on each vertex. An update is muted if the state is 1, otherwise a Bernoulli updating takes place on the vertex and its neighbours.

Sticks and 4-blocks

Next, we will define sticks and then 4-blocks, as well as some special cases. This will allow us to compare our graphical representation to an oriented percolation.

Definition (Stick). Fix $L > 0$. For a vertex $i \in V$ and $k \in \mathbb{Z}_+$, the level- k stick based at i is the set

$$S_{i,k} := \{i\} \times [(k-1)L, kL].$$

Definition (Good stick). Let $A \subseteq \{i\} \cup \mathcal{N}(i)$. A stick $S_{i,k}$ is called A -good if all updates triggered by clock rings on $S_{i,k}$ assign the value 0 to every vertex in A .

Definition (Chain). Let $G = (V, E)$. A sequence $(i_1, \dots, i_n)$ of vertices is called a chain if:

1. $i_j \sim i_{j+1}$ for all $j = 1, \dots, n-1$,
2. the vertices $i_1, \dots, i_n$ are all distinct,
3. for all indices j, k with $|j - k| \geq 3$,

$$\mathcal{N}(i_j) \cap \mathcal{N}(i_k) = \emptyset.$$

Definition (4-block). Let $(i_1, \dots, i_m)$ be a chain and fix $L > 0$. For $k_0 \in \{1, \dots, m-3\}$ and time $t \geq 0$, the 4-block based at k_0 and starting at time t is the space-time set

$$B_{k_0}(t) := \{i_{k_0}, i_{k_0+1}, i_{k_0+2}, i_{k_0+3}\} \times [t, t+L].$$

Definition (Nice 4-block). A 4-block $B_{k_0}(t)$ is called nice if the following hold:

1. For each $j = 0, 1, 2, 3$, the Poisson clock at i_{k_0+j} rings at least once in $[t, t + L]$.
2. The sticks satisfy:

$$\begin{aligned}
i_{k_0}\text{-stick is } \{i_{k_0}, i_{k_0+1}\}\text{-good,} \\
i_{k_0+1}\text{-stick is } \{i_{k_0}, i_{k_0+1}, i_{k_0+2}\}\text{-good,} \\
i_{k_0+2}\text{-stick is } \{i_{k_0+1}, i_{k_0+2}, i_{k_0+3}\}\text{-good,} \\
i_{k_0+3}\text{-stick is } \{i_{k_0+2}, i_{k_0+3}\}\text{-good.}
\end{aligned}$$

3. For any vertex w adjacent to at least one of $i_{k_0}, \dots, i_{k_0+3}$, the stick at w is A_w -good, where

$$A_w := \{i_{k_0}, i_{k_0+1}, i_{k_0+2}, i_{k_0+3}\} \cap \mathcal{N}(w).$$

4. There exist times

$$t \leq t_1 < t_2 < t_3 \leq t + L$$

such that the clocks ring successively at $i_{k_0}, i_{k_0+1}, i_{k_0+2}$ at times t_1, t_2, t_3 , and similarly there exist

$$t \leq s_1 < s_2 < s_3 \leq t + L$$

such that the clocks ring successively at $i_{k_0+3}, i_{k_0+2}, i_{k_0+1}$. This is called occurrence of the propagation event.

Now, due to the spacing between 4-blocks, the neighbourhoods of the "extreme" vertices (i.e. i_{k_0}, i_{k_0+3}) are always disjoint from those of other blocks. Thus, the events that distinct 4-blocks are nice are independent. This makes it straight forward for us to compare to an oriented percolation.

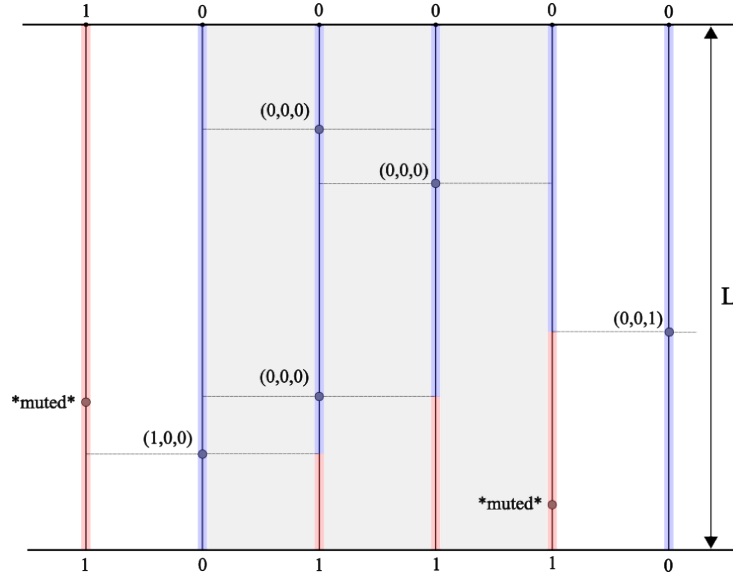


Figure 5.2: Example of nice 4-block in which a zero at one of the extreme sites propagates upwards. The notation $(\cdot, \cdot, \cdot)$ denotes the realizations of the updating Bernoulli random variables.

An important property of these nice 4-blocks discussed in [7] is that zeros propagate upwards. An example of this is seen in figure 5.2. This means that a zero at the beginning time t of a nice block at either extreme vertex i_{k_0} or i_{k_0+3} will lead to zeros at both points i_{k_0} and i_{k_0+3} at the end time $t + L$. The ordering of clock rings in the propagation event and good-stick conditions ensure that a zero at an extreme vertex at the beginning spreads at each such ring and is never overwritten by a one.

Proof of survival

To begin this proof, suppose that we are on the chain $(0, \dots, N-1)$ arranged on the circumference C_N with periodic boundary conditions. Consider the graphical representation and define 4-blocks in space-time, as in figure 5.3, by

$$B_{k,j} = \{3k, 3k+1, 3k+2, 3k+3\} \times [jL, (j+1)L], \text{ for } k, j \in \mathbb{Z} \text{ with } k+j \text{ even,}$$

where spatial indices are taken modulo N .

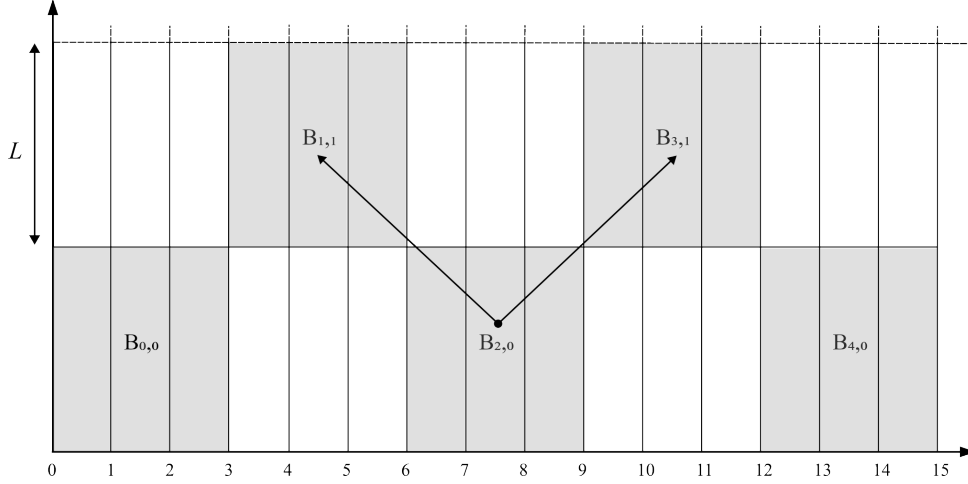


Figure 5.3: Graphical representation on $(0, \dots, N-1)$ partitioned into 4-blocks with arrows representing edges of an oriented percolation.

We make a direct comparison with the oriented percolation model described earlier. We associate each 4-block $B_{k,j}$ with a site $(k, j) \in \mathcal{L}$. A site (k, j) is considered to be open if the corresponding 4-block $B_{k,j}$ is nice, and closed otherwise as shown in figure 5.4. Thus, if a zero in the original process occupies one of the extreme sites of a nice 4-block, then this zero propagates upward through the corresponding open sites in the oriented percolation process. It is then useful to show that the associated oriented percolation process is supercritical for some small $p > 0$. To do so, we will first maximize the probability of a 4-block being nice with respect to L .

We first derive a lower bound for the probability that a 4-block is nice. This will be done by identifying one specific case in which a 4-block is nice where we set starting time to $t = 0$ w.l.o.g. since 4-blocks have an equal probability of being nice over time. Suppose i_{k_0} and i_{k_0+3} both ring at least once in time $[0, L/3]$, while x_{k_0+1} and i_{k_0+2} both ring in $[L/3, 2L/3]$ and in $[2L/3, L]$. If the condition that all sticks are good holds, this 4-block is indeed nice by construction. Thus, its probability of taking place is a suitable lower bound for the probability of a 4-block being nice.

To derive this bound, we first bound from below the probability that a stick is A-good. Let N be the number of rings of the clock on the stick during an interval of length L . Then

$$N \sim \text{Poisson}(L), \quad \mathbb{P}(N = k) = e^{-L} \frac{L^k}{k!}.$$

Suppose A is the set of vertices which must receive zero marks, with $|A| = r$. Then, for $N = k$, the stick is A-good if every one of the k rings produces zeros on all vertices in A . Since each vertex receives a zero with a probability that we will denote by $q = 1 - p$

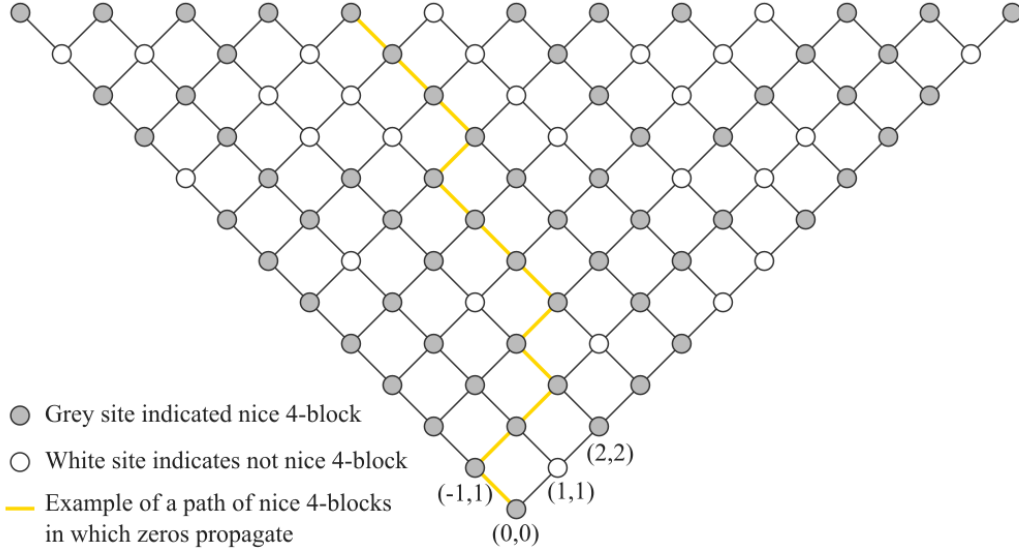


Figure 5.4: Oriented percolation comparison in which each site corresponds to a 4-block.

$$\begin{aligned}
\mathbb{P}(\text{stick is } A\text{-good}) &= \sum_{k=0}^{\infty} \mathbb{P}(N = k) q^{rk} \\
&= \sum_{k=0}^{\infty} e^{-L} \frac{L^k}{k!} q^{rk} \\
&= e^{-L} \sum_{k=0}^{\infty} \frac{(Lq^r)^k}{k!} \\
&= e^{-L} e^{Lq^r} \\
&= e^{-L(1-q^r)}.
\end{aligned}$$

This gives us the following probabilities:

$$\mathbb{P}(i_{k_0}\text{-stick is } \{i_{k_0}, i_{k_0+1}\}\text{-good}) = \mathbb{P}(i_{k_0+3}\text{-stick is } \{i_{k_0+2}, i_{k_0+3}\}\text{-good}) = e^{-L(1-q^2)},$$

$$\mathbb{P}(i_{k_0+1}\text{-stick is } \{i_{k_0}, i_{k_0+1}, i_{k_0+2}\}\text{-good}) = \mathbb{P}(i_{k_0+2}\text{-stick is } \{i_{k_0+1}, i_{k_0+2}, i_{k_0+3}\}\text{-good}) = e^{-L(1-q^3)}.$$

Since there are only two neighbouring sites outside the 4-block, namely i_{k_0-1} and i_{k_0+4} (indices modulo N). Their sticks only need to be good with respect to one site in the block:

$$\mathbb{P}(i_{k_0-1}\text{-stick is } \{i_{k_0}\}\text{-good}) = \mathbb{P}(i_{k_0+4}\text{-stick is } \{i_{k_0+3}\}\text{-good}) = e^{-L(1-q)}.$$

It remains to look at the required ring-time conditions for the propagation event, however these are not independent of the good stick events, so we compute the corresponding joint probabilities.

For the extreme sticks, we require that the stick is good and that there is at least one ring in the interval $[0, L/3]$. Conditioning on the number of rings $N = k$, the ring times are i.i.d. uniform on $[0, L]$, hence the probability that at least one of the k rings lies in $[0, L/3]$ is

$$1 - \left(\frac{2}{3}\right)^k.$$

Therefore,

$$\begin{aligned}
& \mathbb{P}(i_{k_0}\text{-stick is } \{i_{k_0}, i_{k_0+1}\}\text{-good and rings in } [0, L/3]) \\
&= \sum_{k=1}^{\infty} e^{-L} \frac{L^k}{k!} q^{2k} \left(1 - \left(\frac{2}{3} \right)^k \right) \\
&= e^{-L} \left(e^{Lq^2} - e^{2Lq^2/3} \right).
\end{aligned}$$

The same expression holds for the stick based at i_{k_0+3} .

Next, for the middle sticks we require that the stick is good and that there is at least one ring in both intervals $[L/3, 2L/3]$ and $[2L/3, L]$. Conditioning again on $N = k$, the probability that both intervals contain at least one ring equals

$$1 - 2 \left(\frac{2}{3} \right)^k + \left(\frac{1}{3} \right)^k.$$

Hence,

$$\begin{aligned}
& \mathbb{P}(i_{k_0+1}\text{-stick is } \{i_{k_0}, i_{k_0+1}, i_{k_0+2}\}\text{-good} \\
& \quad \text{and rings in both } [L/3, 2L/3] \text{ and } [2L/3, L]) \\
&= \sum_{k=2}^{\infty} e^{-L} \frac{L^k}{k!} q^{3k} \left(1 - 2 \left(\frac{2}{3} \right)^k + \left(\frac{1}{3} \right)^k \right) \\
&= e^{-L} \left(e^{Lq^3} - 2e^{2Lq^3/3} + e^{Lq^3/3} \right).
\end{aligned}$$

The same expression also holds for the stick based at i_{k_0+2} .

Finally, as clock rings are independent on different vertices, we can the contributions of the two extreme sticks, the two middle sticks, and the two neighbouring outside sticks to get

$$\begin{aligned}
\mathbb{P}(4\text{-block is nice}) &\geq e^{-2L(1-q)} \left[e^{-L} \left(e^{Lq^2} - e^{2Lq^2/3} \right) \right]^2 \\
&\quad \cdot \left[e^{-L} \left(e^{Lq^3} - 2e^{2Lq^3/3} + e^{Lq^3/3} \right) \right]^2.
\end{aligned}$$

Rearranging terms yields

$$\begin{aligned}
\mathbb{P}(4\text{-block is nice}) &\geq e^{-6L} \cdot e^{2Lq} \left(e^{Lq^2} - e^{2Lq^2/3} \right)^2 \left(e^{Lq^3} - 2e^{2Lq^3/3} + e^{Lq^3/3} \right)^2 \\
&= e^{\left(\frac{2q^3}{3} + \frac{4q^2}{3} + 2q - 6 \right)L} \left(e^{Lq^2/3} - 1 \right)^2 \left(e^{Lq^3/3} - 1 \right)^4 \\
&=: \mathcal{P}(L, p).
\end{aligned}$$

Next, we will maximize $\mathcal{P}(L, p)$ with respect to L in order to find the closest such bound. Write

$$\mathcal{P}(L, p) = e^{a(q)L} \left(e^{Lq^2/3} - 1 \right)^2 \left(e^{Lq^3/3} - 1 \right)^4, \quad a(q) = \frac{2q^3}{3} + \frac{4q^2}{3} + 2q - 6.$$

Differentiating the natural logarithm with respect to L gives

$$\frac{\partial}{\partial L} \ln \mathcal{P}(L, p) = a(q) + \frac{2q^2}{3} \frac{e^{Lq^2/3}}{e^{Lq^2/3} - 1} + \frac{4q^3}{3} \frac{e^{Lq^3/3}}{e^{Lq^3/3} - 1}.$$

We have that

$$q = 1 - p, \quad q^2 = 1 - 2p + O(p^2), \quad q^3 = 1 - 3p + O(p^2).$$

thus, we obtain the following expressions

$$a(q) = -2 - \frac{20}{3}p + O(p^2),$$

$$\frac{2q^2}{3} + \frac{4q^3}{3} = 2 - \frac{16}{3}p + O(p^2).$$

Using the identity

$$\frac{e^x}{e^x - 1} = 1 + \frac{1}{e^x - 1},$$

we may rewrite

$$\begin{aligned} \frac{\partial}{\partial L} \ln \mathcal{P}(L, p) &= a(q) + \frac{2q^2}{3} \frac{e^{Lq^2/3}}{e^{Lq^2/3} - 1} + \frac{4q^3}{3} \frac{e^{Lq^3/3}}{e^{Lq^3/3} - 1} \\ &= \left(a(q) + \frac{2q^2}{3} + \frac{4q^3}{3} \right) + \frac{2q^2/3}{e^{Lq^2/3} - 1} + \frac{4q^3/3}{e^{Lq^3/3} - 1}. \\ \implies \frac{\partial}{\partial L} \ln \mathcal{P}(L, p) &= -12p + \frac{2q^2/3}{e^{Lq^2/3} - 1} + \frac{4q^3/3}{e^{Lq^3/3} - 1} + O(p^2). \end{aligned}$$

As $p \rightarrow 0$, we have $q \rightarrow 1$, so asymptotically

$$\frac{\partial}{\partial L} \ln \mathcal{P}(L, p) \sim -12p + \frac{2}{e^{L/3} - 1}.$$

Setting this equal to zero gives $e^{L/3} - 1 \sim \frac{1}{6p}$, hence for small p the asymptotically optimal 4-block length is $\hat{L} \sim 3 \ln \frac{1}{6p}$.

Substituting gives

$$\mathcal{P}(\hat{L}, p) = 1 - 36p \ln(p^{-1}) + O(p), \quad p \rightarrow 0.$$

In particular, $\mathcal{P}(\hat{L}, p) \rightarrow 1$ as $p \rightarrow 0$ and hence $\mathbb{P}(\text{4-block is nice}) \rightarrow 1$. Thus, for sufficiently small p , the oriented percolation discussed is indeed supercritical.

Hence, $\mathbb{P}(\text{there exists an infinite oriented open path}) > 0$ and in our discrete Bak-Sneppen model zeros survive when p is sufficiently small.

Lower bound for critical value

We would like to find a bound on p such that this oriented percolation is supercritical and we have survival of zeros.

The maximization of $\mathcal{P}(L, p)$ with respect to L is not analytically solvable, hence why we solved it asymptotically before. Thus, using the rigorous upper bound $\phi_c < 0.72599$ for the critical parameter of oriented percolation, we numerically maximize $\mathcal{P}(L, p)$ with respect to L for each fixed p .

The numerical optimization gives the approximate threshold $p^* \approx 0.001381$ with corresponding maximizing block length $\hat{L}(p^*) \approx 14.4492$. For this value, $\mathcal{P}(\hat{L}(p^*), p^*) \approx 0.7491$.

Thus, the computations suggest that the induced oriented percolation process is supercritical for $p < 0.001381$.

We equivalently have the lower bound

$$p_c > 0.001381.$$

Final proof

From here on we will restrict our analysis of the oriented percolation to the interval $[-N, N]$ for some large $N \in \mathbb{N}$. We take sufficiently small p such that the oriented percolation comparison is supercritical. Denote the height H_k of the k -th consecutive finite cluster after some starting time as the time that the cluster survives for before extinction:

$$H_k := \sup\{n \geq 0 : \xi_n^{(k)} \cap [-N, N] \neq \emptyset\} < \infty,$$

where $\xi_n^{(i)}$ denotes the set of open sites at level n reachable from the k -th finite cluster's starting site. The H_k 's are i.i.d. random variables.

After some time, in the model not constrained to $[-N, N]$, we expect infinite clusters to appear. By the metastable behaviour of our process, in this case we will observe a large cluster that completely fills up the interval $[-N, N]$ and takes a very long time to die. This event takes place with probability $\theta(p)$ and hence, the number of finite clusters we see consecutively from some starting time is distributed as a Geometric random variable $K \sim \text{Geo}(\theta(p))$. Thus, we can define the time between the beginning of a finite cluster and the beginning of the next infinite cluster as $\hat{T}_1 = \sum_{k=1}^K H_k$.

From the beginning of the following infinite cluster we have from Theorem 2.0.6 that the time it takes for it to cover the entire interval $[-N, N]$ is given by $\hat{T}_2 \approx \alpha N$.

After this time, by Theorem 2.0.11, on the interval we will have a positive density of zero sites. As from 2.0.8 this is asymptotically equivalent to the all zero (on $[-N, N]$) starting configuration, we can apply Theorem 2.0.10 to state that the cluster will survive for a long time, notably $\hat{T}_3 \sim \text{Exp}(1)\beta_N$ where $\beta_N \rightarrow \infty$ as $N \rightarrow \infty$.

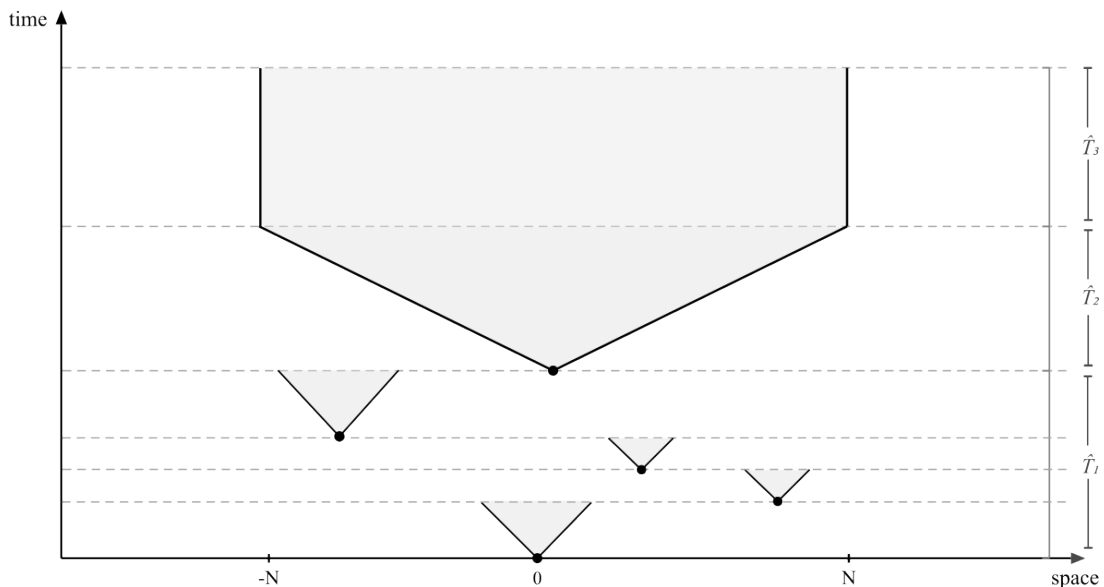


Figure 5.5: Example of the 3 described Space-time phases which continuously repeat in the order $\hat{T}_1$ (finite clusters) $\rightarrow \hat{T}_2$ (growth) $\rightarrow \hat{T}_3$ (long survival)

The 3 events described are shown in figure 5.5. After the extinction of this infinite cluster, the successive events corresponding to times $\hat{T}_1 \rightarrow \hat{T}_2 \rightarrow \hat{T}_3$ repeat for the rest of time. Hence, we will denote $\hat{T}_1^{(i)}, \hat{T}_2^{(i)}, \hat{T}_3^{(i)}$ as the i.i.d. times of the i -th recurrences of each respective event after time $t = 0$.

Now that we know there is a positive density of zeros on $[-N, N]$ throughout each time $\hat{T}_3^{(i)}$,

we will look at the fraction of time we spend in this state. Suppose for some $T \in \mathbb{R}$ we choose randomly $t \in [0, T]$ and define $I = \max\{i : T_3^{(i)} \leq t\}$. Then we compute

$$\begin{aligned} \mathbb{P}(t \in \hat{T}_3^{(i)} \text{ for some } i \in \{1, \dots, I\}) &= \frac{\sum_{i=1}^I \hat{T}_3^{(i)}}{\sum_{i=1}^I (\hat{T}_1^{(i)} + \hat{T}_2^{(i)} + \hat{T}_3^{(i)})} \\ &= \frac{(1/I) \sum_{i=1}^I \hat{T}_3^{(i)}}{(1/I) \sum_{i=1}^I (\hat{T}_1^{(i)} + \hat{T}_2^{(i)} + \hat{T}_3^{(i)})} \\ &\rightarrow \frac{\mathbb{E}[\hat{T}_3]}{\mathbb{E}[\hat{T}_1] + \mathbb{E}[\hat{T}_2] + \mathbb{E}[\hat{T}_3]} \text{ as } T \rightarrow \infty. \end{aligned}$$

where the convergence at the last step follows as a consequence of the law of large numbers. These expectations can be computed as

$$\mathbb{E}[\hat{T}_1] = \mathbb{E}[K] \cdot \mathbb{E}[H_k] = \frac{1}{\theta(p)} \cdot \mathbb{E}[H_k],$$

$$\mathbb{E}[\hat{T}_2] = \alpha N,$$

$$\mathbb{E}[\hat{T}_3] = \beta_N.$$

and thus,

$$\mathbb{P}(t \in \hat{T}_3^{(i)} \text{ for some } i \in \{1, \dots, I\}) \rightarrow \frac{\beta_N}{\frac{1}{\theta} \cdot \mathbb{E}[H_k] + \alpha N + \beta_N} \text{ as } T \rightarrow \infty.$$

From Theorem 2.0.10 we have that

$$\frac{\mathbb{E}[T_N]}{\beta_N} = \mathbb{E}\left[\frac{T_N}{\beta_N}\right] \rightarrow 1 \text{ as } N \rightarrow \infty.$$

i.e. $\beta_N \sim \mathbb{E}[T_N]$ as $N \rightarrow \infty$. We know that with probability $1 - \theta(p)$, T_N is small, and with probability $\theta(p)$, $T_N \geq \alpha(p)N$ (it grows to fill the whole interval). Therefore,

$$\beta_N \sim \mathbb{E}[T_N] > \theta(p) \cdot \alpha N.$$

This, alongside that

$$\mathbb{E}(H_k) = \int_0^\infty \mathbb{P}(H_k > t) dt \leq \int_0^\infty e^{-\epsilon t} dt = \frac{1}{\epsilon} \text{ for some } \epsilon > 0,$$

since by Theorem 2.0.9 $\mathbb{P}(H_k > t) \leq e^{-\epsilon t}$, show that

$$\frac{\beta_N}{\frac{1}{\theta} \cdot \mathbb{E}[H_k] + \alpha N + \beta_N} > \frac{\theta \alpha}{\alpha + \theta \alpha} = \frac{\theta}{1 + \theta} > 0 \text{ for large } N.$$

Therefore in the case as $N \rightarrow \infty$,

$$\mathbb{P}(t \in T_3^{(i)} \text{ for some } i \in \mathbb{N}) > 0,$$

meaning that we spend a portion of our time in this state in which a positive density of sites are contained within an infinite open cluster. Bringing this back to the Bak-Sneppen model:

$$\mathbb{P}((k, j) \in \mathcal{L} \text{ is in a cluster}) = \mathbb{P}(B_{k,j} \text{ is in a path of nice blocks}) > 0$$

$$\implies \lim_{t \rightarrow \infty} \mathbb{P}(X_i(t) = 0) > 0$$

$$\implies \limsup_{N \rightarrow \infty} \pi_{N,p}(X : X_i = 1) < 1.$$

Thus, concluding our proof. □

6. Simulation & numerical results

Numerical investigation of zero survival

To numerically investigate the survival of zeros in the discrete Bak–Sneppen model, we simulated the process on the ring C_N for several system sizes and Bernoulli parameters p .

For each simulation, the quantity

$$Z_t := \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{X_i(t)=0\}}$$

was measured, representing the fraction of vertices in state 0 at time t .

To estimate the stationary zero density, the process was run for a total of 100 cycles, where one cycle corresponds to N update steps and thus the number of iterations is scaled with the number of species. The first 40 cycles were discarded as burn-in i.e. cycles needed to approximately reach the stationary density, and this density was then computed by averaging Z_t over the remaining simulation time.

We considered the values

$$p \in \{0.0005, 0.001, 0.002\}$$

and number of species

$$N \in \{500, 1000, 2000, 5000, 10000\}.$$

The resulting stationary densities of zeros were computed as in Table 6.1.

p	$N = 500$	$N = 1000$	$N = 2000$	$N = 5000$	$N = 10000$
0.0005	0.999470	0.999068	0.999256	0.999286	0.999251
0.001	0.998549	0.998422	0.998364	0.998410	0.998569
0.002	0.996949	0.997502	0.996953	0.996993	0.996957

Table 6.1: Estimated stationary zero densities for several values of p and system sizes N .

The simulations indicate, that for very small values of p the density of zeros remains close to 1. In addition, the estimated stationary densities appear to be stable as N increases.

These numerical results are consistent with the theoretical lower bound for the critical value obtained previously, namely that zeros survive with positive probability for $p < 0.002019$. The results also indicate that the survival of zeros persists as we approach an infinite number of species. This was our main result from the previous section.

Numerical investigation of the supercritical regime

We now numerically investigate the regime of larger values of p , corresponding to the supercritical behaviour of the discrete Bak–Sneppen model, i.e. the regime in which the system converges towards the all-ones configuration.

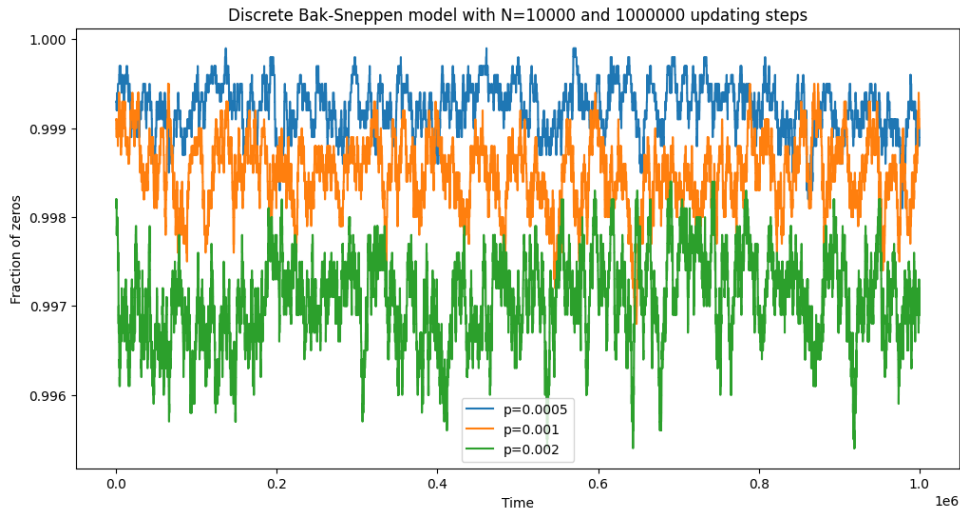


Figure 6.1: Zero survival for small p

This is done as before by averaging the quantity Z_t over 100 cycles with burn-in period of 40 cycles.

We then considered the values

$$p \in \{0.3, 0.35, 0.4, 0.45, 0.5, 0.55\}$$

and number of species as before.

The resulting estimated stationary zero densities are shown in Table 6.2.

p	$N = 500$	$N = 1000$	$N = 2000$	$N = 5000$	$N = 10000$
0.3	0.492620	0.472905	0.477611	0.481452	0.479261
0.35	0.304514	0.319714	0.296516	0.310530	0.319554
0.4	0.019535	0.007724	0.003883	0.001617	0.000851
0.45	0.008628	0.004081	0.001983	0.000797	0.000406
0.5	0.005628	0.002733	0.001345	0.000544	0.000275
0.55	0.004151	0.002090	0.001054	0.000416	0.000208

Table 6.2: Estimated stationary zero densities for several values of p , some of which in the supercritical regime, and different system sizes N .

Based on Table 6.2, for $p = 0.3$ and $p = 0.35$, the stationary zero density seems to remain bounded away from zero as N increases, suggesting that zeros continue to survive as the number of species tends to infinity.

In contrast, for $p \geq 0.4$, as can be seen in Figure 6.2, the stationary zero density decreases rapidly as N increases. In particular, for $p = 0.45, 0.5$, and 0.55 , the density of zeros appears to converge towards 0 as $N \rightarrow \infty$. Equivalently, the stationary density of ones appears to converge towards 1.

These numerical observations are consistent with the rigorous result of (4.2), which states that for $p > 0.45 \dots$, the stationary probability that a vertex is in state 1 converges to 1 as the number of species tends to infinity.

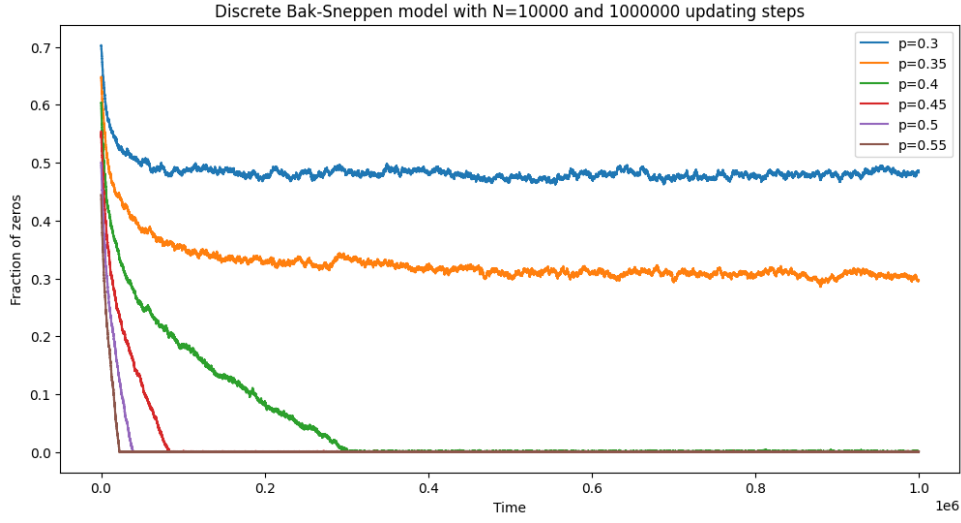


Figure 6.2: Investigation of convergence to the supercritical regime

Estimation of the critical value

The previous section gave an indication that the critical value lies in the interval $p_c \in [0.3, 0.4]$. We now investigate the stationary behaviour for the following

$$p \in \{0.36, 0.365, 0.37, 0.375, 0.38\}$$

for very large $N = 100000$ over 100 cycles.

The behaviour of this system is shown in Figure 6.3. It suggests that $p_c \approx 0.365$, although significantly larger values of N and more updating steps would be required to obtain a conclusive estimate.

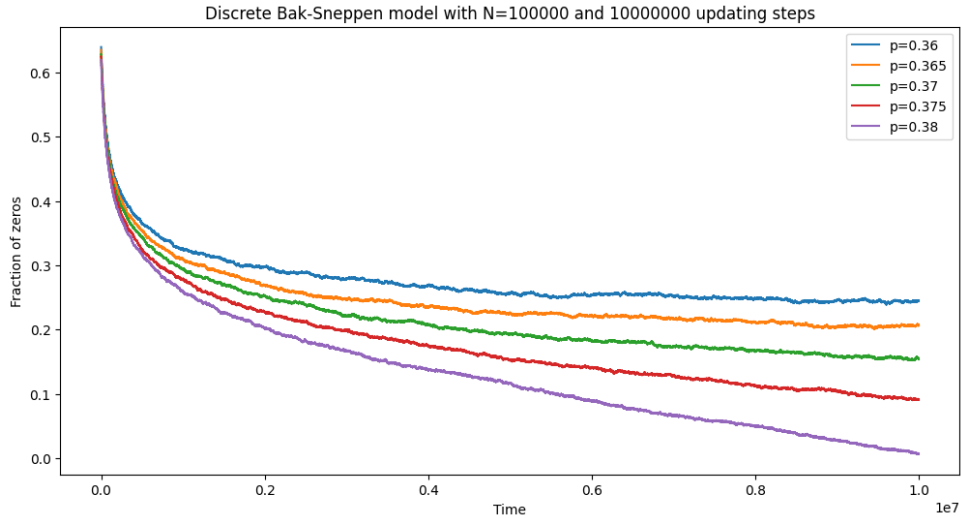


Figure 6.3: Critical Value estimation

7. Conclusion

Summary of Results

In this thesis, we have studied the discrete Bak-Sneppen model on the ring C_N , in which N species are assigned binary fitness values and evolve through a Bernoulli(p) updating rule. The central object of study was the critical parameter p_c , which separates a regime in which zeros survive in the stationary distribution from one in which the system converges to the all-ones configuration.

We presented two main theoretical results, together establishing that p_c is bounded away from both zero and one.

The first result, following Volkov [9], provides a rigorous upper bound on the critical parameter. By introducing the adjusted diameter M_t and the minimal spanning arc of zeros, we showed that for $p > p^* = 0.45\dots$, where p^* is the unique positive real solution of

$$p^5 + 4p^4 + 2p^3 + 3p^2 = 1,$$

the stationary probability that a given vertex has fitness one satisfies $\pi^{(N)}(p) \rightarrow 1$ as $N \rightarrow \infty$. This gives the upper bound $p_c \leq p^* = 0.45\dots$.

The second result, adapted from the approach of Popov and Volkov [7], establishes a lower bound. By constructing a graphical representation of the discrete Bak-Sneppen dynamics in continuous time and partitioning space-time into independent 4-blocks, we related the process to a two-dimensional oriented percolation model. Using the known supercriticality bound $\phi_c < 0.72599$ for oriented percolation [2], we showed that for sufficiently small p , the induced percolation process is supercritical and zeros survive with positive density in the stationary distribution. Numerical optimisation of the 4-block length L to maximise probability of it being nice yielded the lower bound

$$p_c > 0.00138.$$

Finally, we complemented these theoretical results with numerical simulations, which confirmed the survival of zeros for small p and convergence to the all-ones state for $p > p^*$. The simulations further suggest an empirical estimate of $p_c \approx 0.365$.

Discussion of the Bounds

The theoretical bounds established in this thesis leave a substantial gap, with

$$0.00138 < p_c \leq 0.45\dots,$$

while numerical simulations suggest the true critical value lies near $p_c \approx 0.365$. This gap reflects the inherent difficulty of the model but also the limitations of methods used.

The upper bound $p_c \leq 0.45\dots$ is relatively tight, as it arises from a direct drift analysis of the adjusted diameter M_t . The introduction of the adjustment parameter β already provides a meaningful improvement over using the standard arc length D_t , and the bound p^* is determined

precisely by the point at which the drift can no longer be guaranteed to be negative. Sharpening this bound would require either a more refined function M_t or a fundamentally different approach.

The lower bound $p_c > 0.00138$, by contrast, is far from the empirical estimate and is quite conservative. This conservatism stems from several sources of loss in the oriented percolation comparison. The good-stick conditions require that every clock ring within a 4-block produces a zero on all required neighbours, which becomes increasingly unlikely as p grows. Additionally, the lower bound on the probability of a 4-block being nice, $\mathcal{P}(L, p)$, already underestimates the true probability of a 4-block being nice, since only one specific realisation of ring times is accounted for. The result is that the percolation comparison only enters the supercritical regime for very small p , even though in practice the zeros survive for much larger values.

The large gap between the two bounds also reflects a deeper difficulty of the model: the discrete Bak-Sneppen process lacks monotonicity with respect to the natural partial order on configurations, which prevents the use of standard coupling and stochastic domination arguments that would otherwise allow for sharper bounds. This is in contrast to, for example, the oriented percolation, where monotonicity plays a central role in obtaining tight estimates on the critical parameter.

Open Problems

Several questions remain open following the results of this thesis.

Existence and uniqueness of p_c . Throughout this thesis, we have assumed that the critical parameter p_c is well-defined, that is, that $\theta_{BS}(p)$ is non-increasing in p . This is intuitively clear, since a higher probability of assigning fitness one should make it harder for zeros to survive, but this has yet to be proven. Proving rigorously that a unique critical threshold exists would be a significant contribution to the theory of the discrete Bak-Sneppen model.

Sharper bounds on p_c . The bounds established in this thesis leave a large gap, $0.00138 < p_c \leq 0.45$, while simulations suggest $p_c \approx 0.365$. Improving the lower bound is particularly challenging, as any approach based on oriented percolation comparison is limited by the conservatism of the good-stick conditions. A sharper lower bound might require a more refined graphical construction, or a different comparison altogether. On the upper bound side, a more refined function beyond the adjusted diameter M_t could potentially bring the rigorous threshold closer to the empirical estimate.

The classical Bak-Sneppen model. The original Bak-Sneppen model uses continuous fitness values drawn from the uniform distribution on $[0, 1]$, rather than the binary values considered here. Rigorous results for the continuous model are considerably harder to obtain, and the existence of a critical probability in that setting remains an open problem. It could be investigated whether the techniques developed here, in particular the graphical representation and block comparison approach, can be adapted to yield any rigorous statements for the continuous model.

Final Remarks

The discrete Bak-Sneppen model, despite its simple formulation, exhibits interesting and mathematically challenging behaviour. The results of this thesis provide rigorous bounds on the critical parameter p_c , supported by numerical simulations, while the open problems outlined above indicate that much remains to be understood.

Bibliography

- [1] Per Bak and Kim Sneppen. “Punctuated equilibrium and criticality in a simple model of evolution”. In: *Physical Review Letters* (1993).
- [2] Paul Balister, Béla Bollobás, and Alan Stacey. “Upper Bounds for the Critical Probability of Oriented Percolation in Two Dimensions”. In: *Proceedings of the Royal Society of London, Series A* (1993).
- [3] Richard Durrett. “Oriented percolation in two dimensions”. In: *The Annals of Probability* (1984).
- [4] Richard Durrett and Roberto H. Schonmann. “Large deviations for the contact process and two dimensional percolation”. In: *Probability Theory and Related Fields* (1988).
- [5] Richard Durrett and Roberto H. Schonmann. “The contact process on a finite set. II”. In: *The Annals of Probability* (1988).
- [6] James R. Norris. *Markov Chains*. Cambridge University Press, 1997.
- [7] Serguei Popov and Stanislav Volkov. “On extinction and survival of the Bak-Sneppen model on arbitrary graphs”. In: *Preprint* (2025).
- [8] Roberto H. Schonmann. “Metastability for the contact process”. In: *Journal of Statistical Physics* (1985).
- [9] Stanislav Volkov. “Rigorous upper bound for the discrete Bak-Sneppen model”. In: *Journal of Statistical Physics* (2021).